

The combined use of symmetric and asymmetric approaches: partial least squares-structural equation modeling and fuzzy-set qualitative comparative analysis

Use of
symmetric and
asymmetric
approaches

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Abstract

Purpose – This study aims to propose guidelines for the joint use of partial least squares structural equation modeling (PLS-SEM) and fuzzy-set qualitative comparative analysis (fsQCA) to combine symmetric and asymmetric perspectives in model evaluation, in the hospitality and tourism field.

Design/methodology/approach – This study discusses PLS-SEM as a symmetric approach and fsQCA as an asymmetric approach to analyze structural and configurational models. It presents guidelines to conduct an fsQCA based on latent construct scores drawn from PLS-SEM, to assess how configurations of exogenous constructs produce a specific outcome in an endogenous construct.

Findings – This research highlights the advantages of combining PLS-SEM and fsQCA to analyze the causal effects of antecedents (i.e., exogenous constructs) on outcomes (i.e., endogenous constructs). The construct scores extracted from the PLS-SEM analysis of a nomological network of constructs provide accurate input for performing fsQCA to identify the sufficient configurations required to predict the outcome(s). Complementing the assessment of the model's explanatory and predictive power, the fsQCA generates more fine-grained insights into variable relationships, thereby offering the means to reach better managerial conclusions.

Originality/value – The application of PLS-SEM and fsQCA as separate prediction-oriented methods has increased notably in recent years. However, in the absence of clear guidelines, studies applied the methods



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inconsistently, giving researchers little direction on how to best apply PLS-SEM and fsQCA in tandem. To address this concern, this study provides guidelines for the joint use of PLS-SEM and fsQCA.

Keywords Partial least squares structural equation modeling (PLS-SEM), Predictive power, Qualitative comparative analysis (QCA), Fuzzy-set qualitative comparative analysis (fsQCA), Prediction-oriented methods, Symmetric and asymmetric techniques

Paper type Research paper

1. Introduction

In recent years, partial least squares structural equation modeling (PLS-SEM; Wold, 1982) has been applied widely in various disciplines, including hospitality and tourism (Ali *et al.*, 2018; do Valle and Assaker, 2016; Usakli and Kucukergin, 2018), accounting (Nitzl, 2016), family business (Sarstedt *et al.*, 2014), human resource management (Ringle *et al.*, 2020), information systems (Ringle *et al.*, 2012), marketing (Hair *et al.*, 2012b) and strategic management (Hair *et al.*, 2012a). Partial least squares (PLS) is a composite-based approach to structural equation modeling (SEM), which allows the testing of theoretical frameworks from a prediction perspective and the analysis of complex models with latent constructs [for more detailed explications of the use of PLS-SEM, see Hair *et al.* (2019)]. In doing so, PLS-SEM produces mean effects, which quantify the average impact of every single exogenous construct (i.e., the independent variable or antecedent) on an endogenous variable (i.e., the dependent variable; Richter *et al.*, 2020). However, as pointed out by numerous authors in different research fields, such a mean-centered symmetric approach to model estimation paints an incomplete picture of the effects (Dul, 2016; Olya, 2020; Ragin, 2006; Woodside, 2013). To address this concern, researchers have called for the use of asymmetric approaches that process and analyze each individual case instead of the variables' – or in the case of PLS-SEM, the constructs' – mean values (Dul, 2016; Richter *et al.*, 2020; Woodside, 2013). In an asymmetric approach, the aim is to explore how various combinations of independent variables generate specific levels of outcome(s), for different groups of cases (Fiss, 2011). These combinations of predictors can be categorized as sufficient or necessary conditions based on the number and ratio of supported cases (Ragin, 2009; Woodside, 2013).

A particularly prominent approach in this field is Ragin's (1987) fuzzy-set qualitative comparative analysis (fsQCA), which has become the standard tool for asymmetric analyses in regression-based models (Woodside, 2013). Recent years have seen a significant increase in fsQCA applications in various social science disciplines (Farmaki *et al.*, 2021; Kadile and Biraglia, 2020; Mikalef and Pateli, 2017; Pappas *et al.*, 2019; Phung *et al.*, 2020; Xie and Wang, 2020), especially in the field of hospitality and tourism (Olya, 2020; Rasoolimanesh *et al.*, 2020). In this field, researchers have applied fsQCA in different contexts, such as evaluating travelers' willingness to pay (Agag *et al.*, 2020), tourism accommodation services (Pappas and Papatheodorou, 2017), festival management (Gannon *et al.*, 2019; Olya *et al.*, 2020), tourist behavior in green hotels (Olya *et al.*, 2019a), peer-to-peer accommodation (De Canio *et al.*, 2020; Pappas, 2017), medical tourism (Olya and Nia, 2020) and tourist engagement (Rasoolimanesh *et al.*, 2020).

Many of these studies have used fsQCA to analyze the effects between theoretical concepts, measured by means of multi-item constructs. To include the corresponding measures in the fsQCA, researchers typically compute sums of items in a set (i.e., sum scores). This practice is problematic as it ignores the effect of measurement error inherent in the indicators (Sarstedt *et al.*, 2016). By contrast, the individual weighting of the indicators in a PLS-SEM analysis accounts for measurement error, thereby increasing the reliability and validity of the model estimates (Yuan *et al.*, 2020). For example, Hair *et al.* (2017) have shown

that sum scores can produce substantial parameter biases and often lag behind PLS-SEM in terms of statistical power. Apart from these reliability and validity-related concerns, researchers relying on sum scores are unable to determine which indicators have a higher or lower relative importance compared to others in the same set. As PLS-SEM provides this additional information, its use is clearly superior compared to sum scores (Hair *et al.*, 2017).

In light of these concerns, complementing the results of PLS-SEM with an fsQCA is very promising as it combines symmetric and asymmetric perspectives in a single analysis, while accounting for measurement error inherent in the measures. The goal of this combination of methods, or multi-methods approach, is to provide more detailed and nuanced insights into the complex causal relationships between antecedent constructs and selected key target constructs (Olya, 2020; Leischnig *et al.*, 2016). In addition, the combination of PLS-SEM and fsQCA facilitates the assessment of a model's predictive power within the confines of its careful development on the grounds of theory and logic (Chin *et al.*, 2020). The combination of these two prediction-oriented methods can therefore produce more actionable prescriptions as managerial recommendations derived from these models are typically predictive in nature (Hair and Sarstedt, 2021).

Researchers have already started using PLS-SEM and fsQCA in tandem, evidenced by our review of papers published in the top-five hospitality and tourism journals. Specifically, we reviewed all papers published in *Annals of Tourism Research*, *International Journal of Contemporary Hospitality Management*, *International Journal of Hospitality Management*, *Journal of Travel Research* and *Tourism Management* over the five-year period from 2016 to 2020, extracting those that jointly used PLS-SEM and fsQCA. In addition, our review considered the *Journal of Business Research*, which “is currently the major outlet of QCA papers in business” (Dul, 2016, p. 1517). Appendix 1 provides an overview of all papers included in our review.

Our search returned 17 papers that jointly applied PLS-SEM and fsQCA. For example, Bortoluzzi *et al.* (2020) used the methods to investigate the effects of online hotel reviews on hotels strategies and management control systems, whereas Mohamed *et al.* (2020) applied PLS-SEM and fsQCA in tandem to investigate the satisfaction and loyalty of Muslim consumers in a tourism and hospitality context. A detailed analysis of the 17 papers discloses inconsistencies in the application of the methods. Most notably, only 7 out of the 17 papers (41.18%) disclosed the parameter settings of and results reporting in the fsQCA, for example, regarding the type of fsQCA output used (complex vs intermediate vs parsimonious) and the applied threshold to discard specific configurations. Similarly, while PLS-SEM and fsQCA represent prediction-oriented methods that combine symmetric and asymmetric perspectives, only one paper assessed the predictive power of both the PLS-SEM and fsQCA approaches. Finally, all papers were unclear regarding the input used for the fsQCA; that is, whether the analysis draws on construct scores derived from the PLS-SEM analysis or on sum scores of the indicator variables, thereby neglecting measurement error. To summarize, the results of our review suggest that prior research is fraught with confusion regarding the joint application of PLS-SEM and fsQCA.

Addressing this concern, this paper proposes guidelines for the joint application of PLS-SEM and fsQCA. Extending the various empirical papers that have jointly applied PLS-SEM and fsQCA (Bortoluzzi *et al.*, 2020; Mohamed *et al.*, 2020; Tran *et al.*, 2019; Zhang and Zhang, 2019), we offer step-by-step guidance and highlight best practices in the field. We illustrate our guidelines using a well-known model from tourism and hospitality research, which examines the impact of urban residents' attitudes on their perceptions of tourism development. Our overarching aim is to encourage the joint use of PLS-SEM and fsQCA in

tourism and hospitality research, to generate more fine-grained insights into variable relationships, thereby offering the means to derive better managerial conclusions.

2. Fuzzy-set qualitative comparative analysis

2.1 Principles of fuzzy-set qualitative comparative analysis

As a set-theoretic technique, qualitative comparative analysis (QCA; [Ragin, 1987](#)) was initially developed during the 1980s – mainly in political science and historical sociology – as a “macro-comparative approach” to complex social and political conditions, but also as a “small-N” approach involving a small number of cases ([Berg-Schlosser et al., 2009](#), p. 3). However, applications of QCA are not limited to small sample size constellations, but they extend to large data sets in different disciplines ([Berg-Schlosser et al., 2009](#); [Kadile and Biraglia, 2020](#); [Mikalef and Pateli, 2017](#); [Pappas et al., 2019](#)). QCA is used to identify the sufficient and necessary conditions to predict or achieve an outcome by combining certain independent variables ([Ragin, 2006](#)). For example, if there are three independent variables (e.g., X , W and Z) that impact on a dependent variable (Y), QCA facilitates identifying which combinations of these independent variables (i.e., antecedents, ingredients and determinants) lead to an outcome (i.e., dependent variable). If both the occurrence of X and W and the occurrence of X and Z result in Y , these two combinations are sufficient conditions leading to outcome Y . It also means that if X and W or X and Z occur simultaneously, the likelihood that Y will occur is high. These sufficient conditions or combinations are called causal recipes or configurations leading to a specific outcome ([Olya and Gavilyan, 2017](#); [Yu et al., 2016](#)). Moreover, if X is present in all combinations that lead to outcome Y , it means that X is a necessary condition for outcome Y ([Berg-Schlosser et al., 2009](#); [Dul, 2016](#)). To identify these sufficient and necessary conditions, the QCA draws on Boolean algebra ([Ragin, 2006](#)).

To identify the sufficient and necessary conditions using Boolean algebra, the QCA identifies set memberships of observations ([Ragin, 2006](#)). To do so, the original variables are rescaled to values between 0 and 1, whereby 0 refers to full non-membership and 1 to full membership. Methodological research has produced different rescaling approaches ([Schneider and Wagemann, 2012](#)), prominently among which is the fsQCA ([Dul, 2016](#)). The fsQCA rescales the original values of dependent and independent variables into continuous fuzzy values, which fall into a $[0, 1]$ interval. For example, a value of 1 in a seven-point Likert scale translates into a fuzzy value of 0, whereas values of 4 and 7 turn into fuzzy values of 0.5 and 1, respectively ([Olya and Alipour, 2015](#); [Olya and Nia, 2020](#)). Analogously, latent variable scores produced by PLS-SEM can be rescaled into fuzzy values. [Figure 1](#) illustrates the rescaling of (latent) variables into fuzzy values. The fuzzy value of $X_i = 0$ of an

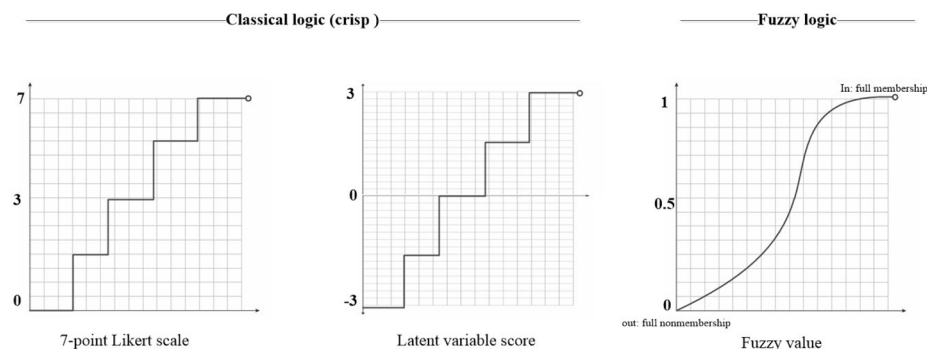


Figure 1.
Interplay between
classical logic and
fuzzy logic

observation i represents no set membership, $0 < X_i < 0.5$, a degree of membership that is more “out” than “in” a set, and $0.5 < X_i < 1$, a degree of membership that is more “in” than “out” a set. The value $X_i = 1$ represents full set membership, while [0.5] is the inflection point at which an observation is neither “in” nor “out” (Mikalef and Pateli, 2017; Ragin, 2009).

To analyze the fuzzy data, researchers need to create a truth table that includes 2^k rows, where k is the number of independent variables and where each row represents a possible combination of all independent variables (i.e., recipe or configuration; Mikalef and Pateli, 2017). For example, if we have three predictors that include X , W and Z , these three variables can occur in eight different conditions or combinations. Table 1 indicates these eight conditions (i.e., recipes or configurations), using Boolean operations to represent the combination of variables (i.e., corresponding vector). Three Boolean operations are used to calculate the configurations. The first negation operation uses the $\sim$ symbol and switches membership of X_i by subtracting the fuzzy-set value from [1], so that $\sim X_i$ equals $[1 - X_i]$. The second operation is logical AND, designated by the $*$ symbol. In this operation, logical AND between two variables takes the minimum fuzzy-set score of two involved variables. The third operation is logical OR, indicated by the $+$ symbol in the configuration, and it takes the maximum fuzzy-set score of the involved variables (Ragin, 2006, 2009). In other words, OR logic not only combines two conditions, but its use allows it to set and represent more conditions. For instance, by combining two first-causal conditions in Table 1 $[(\sim X * \sim W * \sim Z) \text{ OR } (\sim X * \sim W * Z)]$, we establish the new causal condition of $(\sim X * \sim W)$. Therefore, by combining the original 2^k causal conditions through OR logic, $[C(k, k-1) * 2^{k-1}]$ causal conditions can be established which involve $k-1$ predictors. The $C(k, r)$ is called “combination” and refers to the number of selections of r objects from the total available k objects and is calculated based on the following formula: $C(k, r) = \frac{k!}{(k-r)!r!}$. This combination of causal conditions using OR logic will continue to attain single predictor conditions. In this step, the conditions are single predictors (e.g., X , W and Z) and their negations (e.g., $\sim X$, $\sim W$ and $\sim Z$). The total number of conditions is $(3^k - 1)$ (Table 1).

Subsequently, by using each configuration’s consistency and coverage, we can identify the sufficiency or necessity of configurations (i.e., recipes, causal conditions or combinations) for the occurrence of the dependent variable or outcome (Ragin, 2006). The consistency of each configuration (CON _{i}) is calculated based on equation (1) (Ragin, 2006, 2009):

$$\text{Consistency (CON}_i\text{)} = \sum (\min(\text{CON}_i, Y_i)) / \sum (\text{CON}_i) \quad (1)$$

Predictors: independent variables, antecedents and ingredients*			Model: recipe, configuration, combination, algorithm and solution*
X	W	Z	<i>Corresponding vector</i>
0	0	0	$\sim X * \sim W * \sim Z$
0	0	1	$\sim X * \sim W * Z$
0	1	0	$\sim X * W * \sim Z$
0	1	1	$\sim X * W * Z$
1	0	0	$X * \sim W * \sim Z$
1	0	1	$X * \sim W * Z$
1	1	0	$X * W * \sim Z$
1	1	1	$X * W * Z$

Note: *Predictors and model from fsQCA are used with different labels

Table 1.
Truth table for three
predictors

In this consistency equation, CON_i represents the fuzzy-set configuration's membership score of each case number i , Y_i is the fuzzy-set membership outcome of case i and $\min$ indicates the selection of the lowest membership of CON_i and Y_i . If the configuration's membership scores of all cases are lower than the membership scores of their corresponding Y (outcome), the consistency will be 1. If there are only a few cases where the membership scores of CON_i are higher than their corresponding Y_i , the consistency will be slightly less than 1. However, if the configuration's fuzzy-set membership scores are higher than and inconsistent with the corresponding outcome in many cases, the result is one of low consistency. A consistency higher than 0.8 confirms the configuration's sufficiency (i.e., recipe, combination or causal condition) to produce a certain outcome (Mikalef and Pateli, 2017; Ragin, 2009).

Another criterion that confirms a configuration's sufficiency to produce an outcome is its coverage. The coverage is the proportion of a specific configuration with a fuzzy-set membership score higher than 0.5 divided by the total number of outcomes with membership scores greater than 0.5 (Ragin, 2009). Therefore, the coverage value indicates the extent to which a specific configuration is important in producing an outcome. The calculation of the coverage value of a specific configuration CON_i is based on equation (2):

$$\text{Coverage}(CON_i) = \sum (\min(CON_i, Y_i)) / \sum (Y_i) \quad (2)$$

The coverage value can only be calculated after establishing the consistency of a specific configuration (Ragin, 2006). A coverage of more than 0.2 confirms a configuration's sufficiency (Ragin, 2009). Specifically, a coverage of 0.2 implies that 20% of an outcome with a fuzzy-set membership score higher than 0.5 is produced by the configuration.

The coverage calculation of causal conditions in fsQCA can be used to assess the causal conditions' necessity to produce an outcome (Ragin, 2009). If a causal condition's (i.e., configuration or recipe) coverage value is close to 1, it means that this condition is necessary to produce an outcome. Therefore, a configuration consistency and coverage greater than 0.9 indicate the necessary condition to produce an outcome (Ragin, 2006, 2009).

The XY plot is an alternative way of schematically depicting a configuration's consistency and coverage. In the XY plot, the X -axis represents the configuration or causal recipe, whereas the Y -axis represents the outcome condition (or its negation). If X is lower than Y in all cases in a fuzzy-set XY plot, so that all points are located in the upper-left box, the consistency [see equation (1)] is close to 1. However, consistency decreases if, in a few cases, X is higher than Y , and these points are located in the lower-right box. The XY plot both shows and highlights acceptable coverage [see equation (2)], based on the location of points. In addition, the XY plot can be used to test the configuration's predictive power. To do this, we randomly divide the data into two subsamples, perform fsQCA for subsample 1 and draw an XY plot for sufficient configurations, and thereafter draw an XY plot and calculate consistency and coverage by using subsample 2 for sufficient configurations from subsample 1. The out-of-sample predictive power of the model using fsQCA is confirmed if the XY plots for sufficient configurations based on subsample 2 have a consistency and coverage similar to the sufficient configurations drawn from subsample 1 (Olya and Al-ansi, 2018; Olya and Altinay, 2016).

2.2 Combining fuzzy-set qualitative comparative analysis with partial least squares structural equation modeling

In its original form, fsQCA is restricted to the processing of observed variables, such as investments or sales (Ragin, 2009). Theoretical concepts, which are "abstract, unobservable

properties or attributes of a social unit of entity” (Bagozzi and Philipps, 1982, p. 465), can only be considered by creating sum scores (or averages) of the indicators used to operationalize the corresponding constructs. This practice is problematic as it ignores the attenuating effect of measurement error. Numerous studies have shown that the failure to correct for measurement error produces a combination of under and over-estimation in the relationships among constructs in a larger nomological network (Cole and Preacher, 2014; Hair *et al.*, 2017; Yuan *et al.*, 2020). By contrast, SEM methods permit the elimination of measurement error in the analyses (Cole and Preacher, 2014).

To estimate structural equation models, researchers often revert to covariance-based SEM (CB-SEM), which estimates the model parameters so that the difference between the empirical and model-implied covariance matrices is minimized (Jöreskog, 1973). In doing so, the scores of the constructs representing the theoretical concepts are neither needed, nor explicitly estimated (Sarstedt *et al.*, 2016). While it is possible to estimate construct scores within a CB-SEM framework, these estimated scores are not unique (Rigdon *et al.*, 2019). An infinite number of different sets of latent variable scores that equally fit the model are possible (Guttman, 1955). This characteristic, known as factor (score) indeterminacy, renders meaningless any follow-up assessment using the construct scores as input (Hair *et al.*, 2011) and hinders the assessment of the model’s out-of-sample predictive power (Becker *et al.*, 2013).

By contrast, a major advantage of composite-based PLS-SEM is that it always produces a single, specific (i.e., determinate) composite score for each case, once the weights are established. In estimating the indicator weights, the PLS-SEM algorithm considers the indicators’ individual reliabilities and their explanatory power in the model’s downstream relationships – see, for example, Tenenhaus *et al.* (2005) for a detailed description of the PLS-SEM algorithm and its properties. Using these construct scores as input, PLS-SEM applies ordinary least squares regression with the objective of minimizing the residual variance of the endogenous construct scores. In doing so, PLS-SEM follows a symmetric perspective as it produces average effects. However, by subjecting the construct scores, which have been derived in the context of a theoretical model, to an fsQCA, the scope of the analysis can be extended to an asymmetric perspective, which contrasts how a combination of predictor construct scores yield a certain outcome in the endogenous construct scores. As the fsQCA is descriptive in nature, this tandem analysis has no effect on the properties of the PLS-SEM analysis or vice versa. Actually, similar approaches to the processing of construct scores have long been established in a PLS-SEM context. For example, the importance–performance map analysis (Ringle and Sarstedt, 2016) considers the average predictor construct scores as indicative of their performance and contrasts them with the constructs’ total effects. Similarly, finite mixture-based latent class analysis in PLS-SEM relies on construct scores from a standard PLS-SEM analysis as input for a series of mixture regressions to capture unobserved heterogeneity in a PLS path model (Hahn *et al.*, 2002; Sarstedt *et al.*, 2011).

3. Model assessment using partial least squares structural equation modeling and fuzzy-set qualitative comparative analysis

The combination of PLS-SEM and fsQCA for purposes of model assessment follows the procedure shown in Figure 2. In Step 1, PLS-SEM is used to estimate and evaluate the measurement models and the structural model by applying well-established reliability and validity criteria (Hair *et al.*, 2019). After establishing the reliability and validity of the construct measures, the structural model assessment focuses on its explanatory and predictive power. The assessment of the model’s predictive power, in particular, builds on

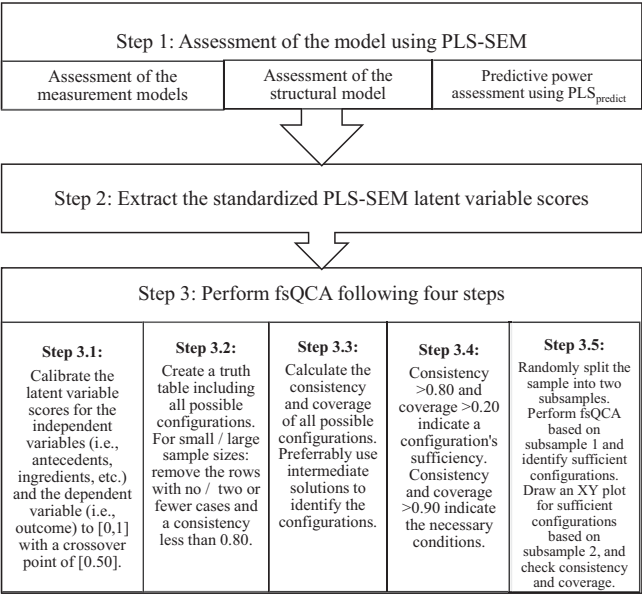


Figure 2.
Model-assessment
procedure using PLS-
SEM and fsQCA

Shmueli *et al.*'s (2016) $PLS_{predict}$ procedure (Shmueli *et al.*, 2019; Danks and Ray, 2018). $PLS_{predict}$ applies k -fold cross-validation to the data set by randomly splitting it into k equally sized subsets of data. The procedure then combines $k-1$ subsets into a training sample, which is used to estimate the model parameters. In the next step, the model estimates are used to generate case-level predictions for all observations in the omitted subset (i.e., the holdout sample). This process is repeated until each subset has served as a holdout sample. Prediction statistics then allow the quantification of the degree of prediction error and an assessment of the model's predictive power *vis-à-vis* a naïve benchmark model. Researchers can also compare alternative model configurations that emerge from different theories or contexts. To compare these models, they should use model selection criteria such as Schwarz's (1978) Bayesian information criterion and Geweke and Meese's (1981) criterion. These criteria are known to achieve a sound trade-off between model fit and predictive power (Sharma *et al.*, 2019, 2020), and they can also be used to compute the relative likelihood of models, given the data and set of alternative models (Danks *et al.*, 2020). In addition, Liengard *et al.*'s (2020) cross-validated predictive ability test enables researchers to determine if an alternative model has a significantly better predictive power than an established model. In Step 2, researchers extract the latent variable scores from the PLS-SEM analysis, which serve as an input for the subsequent fsQCA. The construct scores are computed as linear combinations of the indicator weights, which consider each indicators' measurement error and explanatory power.

In Step 3, fsQCA is performed by means of five sub-steps. Step 3.1 requires calibrating the standardized latent variable scores between -3 (i.e., no set membership) and 3 (full set membership), whereby 0 is the crossover point. Step 3.2 creates a truth table that includes 2^k rows, with k being the number of independent variables and each row representing a possible combination of all the independent variables (i.e., recipe or configuration; Mikalef and Pateli, 2017). Boolean algebra is used to reduce the number of conditions listed in truth

table. In refining the truth tables, [Ragin \(2009\)](#) suggests deleting rows with no cases. In addition, configurations with a consistency of <0.8 should be deleted to ensure that the sufficient and necessary configurations exhibit a satisfactory quality. While these thresholds apply to analyses that draw on relatively small sample sizes of 150 or less, it is possible to consider different thresholds for larger sample sizes. For example, for larger sample sizes, [Fiss \(2011\)](#) suggests deleting rows with two cases or less. Step 3.3 calculates each configuration's consistency and coverage, allowing the researcher to identify the sufficiency or necessity of the configurations (i.e., recipes, causal conditions or combinations). A consistency value higher than 0.8 confirms a configuration's consistency when producing an outcome ([Mikalef and Pateli, 2017](#); [Ragin, 2009](#)). Researchers can calculate the coverage value after establishing a specific configuration's consistency ([Ragin, 2006](#)). The coverage value should be higher than 0.2 to confirm the configuration's sufficiency ([Ragin, 2006, 2009](#)). Using fsQCA, a configuration or single condition (i.e., independent variable) is necessary for an outcome if both consistency and coverage are greater than 0.9 ([Dul, 2016](#); [Ragin, 2009](#)). In Step 3.4, researchers identify the consistency and coverage based on the given thresholds. Researchers need to decide which types of fsQCA outputs (complex, intermediate or parsimonious) should be reported. In a complex output, all the possible configurations leading to the given outcome are calculated using Quine–McCluskey algorithm. While the resulting number of recipes can be large, such an output is appropriate in sensitive application areas, for example in medical tourism ([Olya and Nia, 2020](#)). Because of the solution's complexity, it is not a straightforward task to interpret all complex configurations and, accordingly, derive implications from the analysis. For this reason, researchers typically recommend using intermediate or parsimonious outputs. While a parsimonious output only returns essential configurations, the intermediate variant strikes a balance between completeness and interpretability by focusing on the salient configurations and should, therefore, be preferred ([Olya, 2020](#)). However, interpreting intermediate outputs requires researchers to have sufficient background knowledge of the research field. This enables them to refine all conditions by including core antecedents and compromise peripheral antecedents.

In Step 3.5, the model's predictive power is assessed by using an out-of-sample approach. To do so, researchers first need to randomly split the data set into two subsamples and run the fsQCA on subsample 1 to identify the sufficient configurations. Using the identified sufficient configurations of subsample 1 as input, they must then draw an XY plot and calculate the consistency and coverage based on the data of subsample 2. If subsample 2's XY plot indicates an acceptable and similar consistency and coverage for sufficient configurations identified from subsample 1, the results confirm the predictive power of our model ([Olya and Al-ansi, 2018](#); [Olya and Altinay, 2016](#)). In the final step, the predictive power assessment in fsQCA should be compared to the results from PLS_{predict}.

4. Illustrative example

We use [Rasoolimanesh et al.'s \(2017\)](#) model and data to illustrate the proposed guidelines on the joint application of PLS-SEM and fsQCA. These authors hypothesized the effects of community attachment (CA), environmental attitude (EAT), economic gain (EG) and community involvement (CINV) on residents' positive perceptions (PP) and negative perceptions toward tourism development. Our illustration draws on the data obtained from the urban residents of the George Town World Heritage Site, Malaysia (N = 410). As the fsQCA focuses on a single target construct and for reasons of clarity, we apply a slightly modified and improved model by only considering PP as an endogenous construct ([Figure 3](#)). However, because the model estimates correspond to those reported in

Rasoolimanesh *et al.* (2017), this model modification does not impact the implications from comparing the fsQCA results with the PLS-SEM reported in the original study. To estimate the PLS path model, we rely on the statistical software SmartPLS 3 (Ringle *et al.*, 2015). The fsQCA analysis was carried out using fsQCA 3.0 (Ragin, 2018).

Estimating the PLS path model shows that all indicator loadings are higher than 0.7, thereby supporting their reliability (Hair *et al.*, 2021). Moreover, we confirm the internal consistency reliability of all constructs as their ρ_A reliability results fall in the acceptable level between 0.7 and 0.95 (Ali *et al.*, 2018; Hair *et al.*, 2019). In addition, the results show that the average variance extracted is greater than 0.5, indicative of acceptable convergent validity for all construct measures (Sarstedt *et al.*, 2017). To establish discriminant validity, we used the heterotrait–monotrait ratio of correlations (HTMT) criterion (Henseler *et al.*, 2015). Based on the bootstrapping results of 10,000 subsamples and the percentile approach, we confirm that the HTMT values of all constructs are significantly ($p < 0.05$) lower than the conservative threshold of 0.85, thereby establishing discriminant validity (Franke and Sarstedt, 2019).

To assess the structural model, we considered the path coefficients' significance, again based on bootstrapping, their f^2 effect sizes and the endogenous constructs' R^2 values (Ali *et al.*, 2018; Hair *et al.*, 2019). Moreover, we assessed the model's predictive power by using the PLS_{predict} procedure (Shmueli *et al.*, 2016, 2019). The structural model's path coefficients range between 0.136 and 0.355, indicating that CINV has the strongest effect on the target construct PP (Table 2). All structural coefficients are statistically significant ($p < 0.05$), thus supporting the hypothesized positive relationships of CA, EAT, EG and CINV on PP. The f^2 effect sizes of all four relationships in the structural model range between 0.022 and 0.139 and, therefore, are weak to moderate. With an R^2 value of 0.350, the PP construct's explained variance is satisfactory, considering the model's complexity.

To assess the model's predictive power with regard to the PP construct, we used the PLS_{predict} procedure with ten folds and ten repetitions (Shmueli *et al.*, 2019). We first assessed whether the PLS path model outperforms the training samples' indicator means as

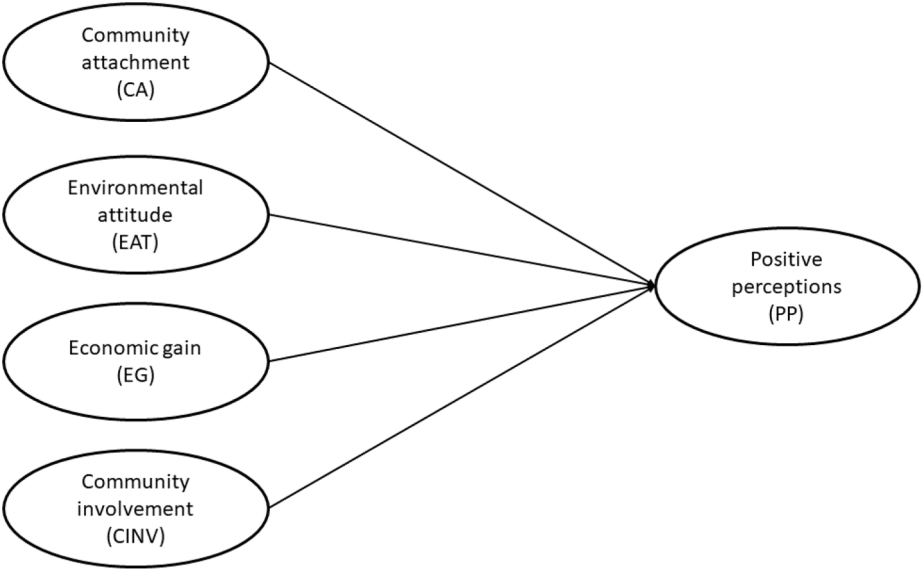


Figure 3.
Conceptual model

evidenced in Q^2_{predict} larger than zero. Next, we compared the root mean squared error (RMSE) generated by the PLS-SEM-based estimates with those of a linear benchmark model (Hair *et al.*, 2020). Table 3 presents the PLS_{predict} results. We find that the Q^2_{predict} values for all the PP construct's indicators are greater than zero. In addition, the results show that the RMSE values for all the PP construct's indicators are lower for PLS-SEM than for the linear model. Collectively, these results indicate that the PLS path model has a high predictive power for the PP construct.

Thereafter, we used PLS-SEM's latent variable scores to perform the fsQCA (Step 2). First, the latent variable scores were calibrated (Step 3.1) for the independent variables and the dependent variable in the framework. In fsQCA, the independent variables are called antecedents or ingredients (i.e., CA, EAT, EG and CINV), while the dependent variable represents the fsQCA's outcome (i.e., residents' PP toward tourism development). Next (Step 3.2), we created a fuzzy-set truth table that included all possible configurations of antecedents. Table 4, from which we removed the alternative with less than three samples, presents the truth tables for the model's outcome. This truth table for the PP outcome shows that by involving all the antecedents (i.e., ingredients), we can only attain a >0.8 consistency if all the antecedents set full membership. However, it is necessary to check the consistency and coverage of all the recipes involving different numbers of antecedents (Step 3.3).

Table 5 presents the sufficient configurations (i.e., recipes) with acceptable consistency (>0.8) and coverage (>0.2) to achieve high and low levels of PP (Step 3.4). The fsQCA results in Table 5 show that whereas $CA*EAT$, $EAT*\sim CINV$, $\sim EAT*EG*CINV$ and $CA*EG*CINV$ generate a high level of PP toward tourism development, $\sim CA*\sim EAT$, $\sim CINV*EAT$, $CINV*EG*\sim EAT$ and $CA*\sim EG*EAT$, by contrast, only generate a low level of PP. That is, a group of residents shows PP toward tourism development if they are attached to a community and have a positive environmental attitude ($CA*EAT$). Another group of residents, although their environmental attitude is not positive, shows PP toward tourism development because they are involved in a community and perceive economic

Table 2.

Path coefficients and
their significance

Hypothesis	Path coefficient	95% bootstrap confidence interval	Effect size (f^2)	Significant at 5%?
CA $\rightarrow$ PP	0.158	[0.033, 0.275]	0.022	Yes
EAT $\rightarrow$ PP	0.147	[0.040, 0.250]	0.023	Yes
EG $\rightarrow$ PP	0.136	[0.066, 0.207]	0.024	Yes
CINV $\rightarrow$ PP	0.355	[0.258, 0.451]	0.139	Yes

Notes: CA: community attachment; EAT: environmental attitude; EG: economic gain; CINV: community involvement; PP: positive perceptions

Table 3.

Results of predictive
power assessment
using PLS_{predict}

Items	Q^2_{predict}	RMSE	
		PLS-SEM	Linear model
PP1	0.263	0.700	0.713
PP2	0.231	0.674	0.694
PP3	0.278	0.676	0.684
PP4	0.222	0.717	0.728

Note: PP: positive perceptions

Table 4.
Truth table for the
configurations to
predict the PP
construct (outcome:
PP)

CA	EAT	EG	CINV	Number	PP	Consistency
1	1	1	1	78	1	0.819
1	0	1	1	18	1	0.629
1	1	0	1	34	1	0.605
1	1	1	0	13	1	0.537
0	0	1	1	33	1	0.378
0	1	1	0	19	1	0.376
0	0	0	1	23	1	0.358
1	1	0	0	28	1	0.351
0	0	1	0	28	1	0.276
0	1	0	0	26	1	0.269
0	0	0	0	41	1	0.185

Notes: CA: community attachment; EAT: environmental attitude; EG: economic gain; CINV: community involvement; PP: positive perceptions

Table 5.
Sufficient
configurations of PP
toward tourism
development

Configurations	Raw coverage	Unique coverage	Consistency
<i>Configurations for high level of PP</i>			
$PP = f(CA, EAT, EG, CINV)$			
CA*EAT	0.753	0.035	0.884
EAT*~CINV	0.622	0.001	0.852
~EAT*EG*CINV	0.575	0.002	0.913
CA*EG*CINV	0.678	0.011	0.933
Solution coverage: 0.929			
Solution consistency: 0.745			
<i>Configurations for low level of PP</i>			
$~PP = f(CA, EAT, EG, CINV)$			
~CA*~EAT	0.759	0.119	0.894
EAT*~CINV	0.671	0.054	0.894
~EAT*EG*CINV	0.587	0.019	0.905
CA*~EG*EAT	0.597	0.014	0.892
Solution coverage: 0.910			
Solution consistency: 0.811			
Notes: CA: community attachment; EAT: environmental attitude; EG: economic gain; CINV: community involvement; PP: positive perceptions			

benefit (~EAT*EG*CINV). By contrast, the PLS-SEM results suggest that all the antecedents have positive effects on residents' PP toward tourism development. However, the fsQCA results obtained by analyzing all the individuals provide further differentiated results. According to the fsQCA results, if both the CA and the EAT are high, the residents' perceptions will be high for both low and high levels of EG and CINV. This result is consistent with the configurations to generate a low level of PP toward tourism development. The result demonstrates that the ~CA*~EAT is a sufficient configuration to generate a low level of PP. However, among the different residents' groups, the fsQCA results generate both high and low PP levels toward tourism development. These results indicate the presence of heterogeneity among the residents. In addition, the fsQCA results show that a high level of CA, EG and CINV is sufficient to generate a high level of PP,

whereas a high level of CA and EAT with a low level of EG is sufficient to generate a low level of PP toward tourism development.

The fsQCA results for sufficient configurations to generate high and low levels of PP demonstrate more complexities when compared with the results of PLS-SEM as a symmetric approach. The necessity analysis' results do not include a consistency and coverage value greater than 0.9 for any of the configurations or single conditions, indicating the absence of the necessary conditions to generate PP.

In the next step, we assess the fsQCA solution's predictive power (Step 3.5). For this purpose, we randomly split the data set into two subsamples. We used subsample 1 to identify the sufficient configurations with a consistency greater than 0.8 and a coverage greater than 0.2. Table 6 shows three sufficient configurations, based on subsample 1, which generate high levels of PP toward tourism development.

Thereafter, we used subsample 2 to draw an XY plot and to calculate consistency and coverage for the extracted sufficient configurations from subsample 1. Figure 4 depicts the results. Subsample 2's results indicate an acceptable level of consistency and coverage for all sufficient configurations to predict PP from subsample 2, confirming the model's predictive power for PP. For the PP construct, the predictive power results using fsQCA are consistent with the results of PLS_{predicts}, confirming the acceptable predictive power of the model that combines the use of PLS-SEM and fsQCA approaches.

Configurations	Raw coverage	Unique coverage	Consistency
<i>Configurations for a high level of PP</i>			
$PP = f(CA, EAT, EG, CINV)$			
CA*EAT*CINV	0.708	0.164	0.927
EAT*~EG*~CINV	0.543	0.000	0.840
CA*~EG*EAT	0.568	0.004	0.871
Solution coverage: 0.890			
Solution consistency: 0.738			

Table 6.
Sufficient
configurations of PP
from subsample 1

Notes: CA: community attachment; EAT: environmental attitude; EG: economic gain; CINV: community involvement; PP: positive perceptions

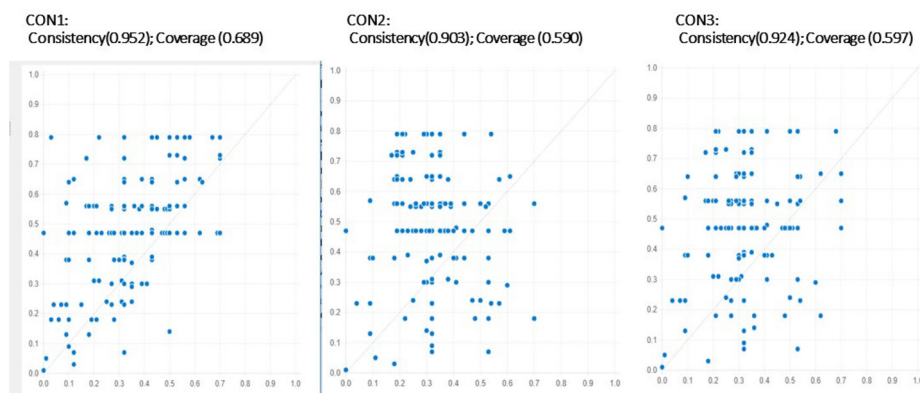


Figure 4.
The XY plots for
sufficient
configurations to
predict PP based on
subsample 2

5. Discussion and conclusions

5.1 Conclusions

Recent research has emphasized the potential of combining the symmetric PLS-SEM approach to model assessment – prevalent in tourism and hospitality research – with an asymmetric perspective using fsQCA, which considers how different configurations of exogenous constructs exert a differentiated effect on an endogenous construct (Bortoluzzi *et al.*, 2020; Elbaz *et al.*, 2018; Mohamed *et al.*, 2020; Tran *et al.*, 2019; Zhang and Zhang, 2019). The goal of this combination of methods, or multi-methods approach, is to provide more detailed and nuanced insights into the complex causal relationships between antecedent constructs and selected key target constructs (Olya, 2020; Leischnig *et al.*, 2016).

Our review of prior studies published in the top-five hospitality and tourism journals (*Annals of Tourism Research*, *International Journal of Contemporary Hospitality Management*, *International Journal of Hospitality Management*, *Journal of Travel Research* and *Tourism Management*) and in the primary outlet for QCA in general management (*Journal of Business Research*), however, shows that authors inconsistently apply the two methods, thus providing little guidance to researchers on how to best combine the methods in their model assessment. Addressing this concern, we present guidelines for the joint use of PLS-SEM and fsQCA. As a result, researchers can use the methods in combination to validate their models or to identify and explore more beneficial causal relationship variations and heterogeneity.

5.2 Theoretical implications

While PLS-SEM and fsQCA are prediction oriented, they offer different perspectives on the model estimates (symmetric vs asymmetric). These complementary perspectives offer more fine-grained insights into variable relationships, thereby providing the means to derive better managerial conclusions.

The application of PLS-SEM is particularly promising as its estimates explicitly account for the measurement error inherent in the indicators and generate construct scores through nomological relationships (Sarstedt *et al.*, 2020b). This sets the method apart from sum scores of multi-item measures, which have routinely been used in applications of fsQCA and related methods – even in seminal presentations of the methods (Dul, 2016). Simultaneously, fsQCA expands and strengthens the PLS-SEM results at an individual level by identifying all possible sufficient combinations of antecedents that generate an outcome.

5.3 Practical implications

The combination of fsQCA and PLS-SEM allows researchers to engage in a more robust assessment of their models' predictive power from both symmetric and asymmetric perspectives. By providing clear guidelines on how to best apply PLS-SEM and fsQCA in tandem, our study offers researchers with the suitable means to analyze their models from these two complementary perspectives.

Our illustrative example demonstrates the value added of complementing the strictly symmetric PLS-SEM method with the asymmetric fsQCA approach. In addition, enriching PLS-SEM analyses with fsQCA offers unique insights into model relationship, which otherwise typically remain undetected. Specifically, the fsQCA results offer novel means to detect nonlinearities (Olya and Altinay, 2016; Sarstedt *et al.*, 2020b) and unobserved heterogeneity in the model (Olya and Gavilyan, 2017; Schlittgen *et al.*, 2016). As such the method complements the traditional PLS-SEM-based latent class analysis methods, whose use has gained considerable traction in recent years (Becker *et al.*, 2013; Olya and Al-Ansi, 2018; Ringle *et al.*, 2013).

5.4 Limitations and future research

Our research offers several avenues for future research. For example, while our empirical illustration draws on standardized construct scores as input for the fsQCA, future research should evaluate the impact of using unstandardized scores, as sometimes used in, for example, PLS-SEM-based impact-performance map analyses (Ringle and Sarstedt, 2016). Prior research has highlighted that the indicator weighting of the PLS-SEM algorithm produces more accurate parameter estimates compared to sum scores (Hair *et al.*, 2017). Future research should extend this perspective to the predictive power assessment in fsQCA. Particularly in constellations with similar indicator weights and small sample sizes, the sum scores approach can be expected to match the fsQCA results (Rigdon, 2012). Future research should also identify thresholds for removing configurations from the truth table for specific research situations. Previous studies recommended a threshold of 1 or 2 for small samples (Ali *et al.*, 2016; Ragin, 2009; Tran *et al.*, 2019). But for larger sample sizes, reliable indications are also needed to guide users. Relatedly, future research should be conducted to identify the most appropriate consistency and coverage thresholds when using fsQCA in combination with PLS-SEM.

PLS-SEM is a powerful method for testing models that involve mediators and moderators (Sarstedt *et al.*, 2020a). While fsQCA can test moderator models by either exploring combinations of predictors or splitting samples into groups (Olya *et al.*, 2019b; Rasoolimanesh *et al.*, 2020), future research should explore means to accommodate mediator relationships. Finally, another area for future research is the development of new criteria to examine and assess the fsQCA solution's predictive power. For example, cross-validation could be used to extend the simple holdout sample-based procedure commonly used in fsQCA applications.

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Appendix

Use of
symmetric and
asymmetric
approaches

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Paper title	Authors	Journal name	Publication year
Effect of online hotel reviews on the relationship between defender and prospector strategies and management controls	Bortoluzzi, D.A., Lunkes, R.J., dos Santos, E.A. and Mendes, A.C.A	<i>International Journal of Contemporary Hospitality Management</i>	2020
Stimulating satisfaction and loyalty: transformative behaviour and Muslim consumers	Mohamed, N., Taheri, B., Farmaki, A., Olya, H. and Gannon, M.J.	<i>International Journal of Contemporary Hospitality Management</i>	2020
E-satisfaction and continuance intention: The moderator role of online ratings	Tran, L.T.T., Pham, L.M. T. and Le, L.T.	<i>International Journal of Hospitality Management</i>	2019
Understanding the influence of airport servicescape on traveler dissatisfaction and misbehavior	Taheri, B., Olya, H., Ali, F. and Gannon, M.J.	<i>Journal of Travel Research</i>	2020
Nepotism, employees' competencies and firm performance in the tourism sector: a dual multivariate and qualitative comparative analysis approach	Elbaz, A.M., Haddoud, M. Y. and Shehawy, Y.M.	<i>Tourism Management</i>	2018
Exploring the impact of big data analytics capabilities on business model innovation: The mediating role of entrepreneurial orientation	Ciampi, F., Demi, S., Magrini, A., Marzi, G. and Papa, A.	<i>Journal of Business Research</i>	2020
Antecedents of innovative performance: Findings from PLS-SEM and fuzzy sets (fsQCA)	Kaya, B., Abubakar, A.M., Behraves, E., Yildiz, H. and Mert, I.S.	<i>Journal of Business Research</i>	2020
Knowledge management capabilities and organizational risk-taking for business model innovation in SMEs	Hock-Doeppen, M., Clauss, T., Kraus, S. and Cheng, C.F.	<i>Journal of Business Research</i>	2020
A mixed methods UTAUT2-based approach to assess mobile health adoption	Duarte, P. and Pinho, J.C.	<i>Journal of Business Research</i>	2019
The adoption of management control systems by start-ups: Internal factors and context as determinants	Crespo, N.F., Rodrigues, R., Samagaio, A. and Silva, G.M.	<i>Journal of Business Research</i>	2019
(continued)			

Table A1.
List of reviewed
papers

Paper title	Authors	Journal name	Publication year
Examining learning strategies, creativity, and innovation at SMEs using fuzzy set qualitative comparative analysis and PLS path modeling	Valaei, N., Rezaei, S. and Ismail, W.K. W.	<i>Journal of Business Research</i>	2017
The mediating role of competitive strategy in international entrepreneurial orientation	Hernández-Perlines, F., Moreno-García, J. and Yáñez-Araque, B.	<i>Journal of Business Research</i>	2016
Direct and configurational paths of absorptive capacity and organizational innovation to successful organizational performance	Ali, M., Kan, K.A. S. and Sarstedt, M.	<i>Journal of Business Research</i>	2016
The effect of public policies on entrepreneurial activity and economic growth	Castaño, M.S., Méndez, M. T. and Galindo, M.Á.	<i>Journal of Business Research</i>	2016
Net versus combinatory effects of firm and industry antecedents of sales growth	Leischnig, A., Henneberg, S.C. and Thornton, S.C.	<i>Journal of Business Research</i>	2016
Rethinking organizational learning orientation on radical and incremental innovation in high-tech firms	Sheng, M.L. and Chien, I.	<i>Journal of Business Research</i>	2016
Using fuzzy-set qualitative comparative analysis to develop an absorptive capacity-based view of training	Hernández-Perlines, F., Moreno-García, J. and Yáñez-Araque, B.	<i>Journal of Business Research</i>	2016

Table A1.

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